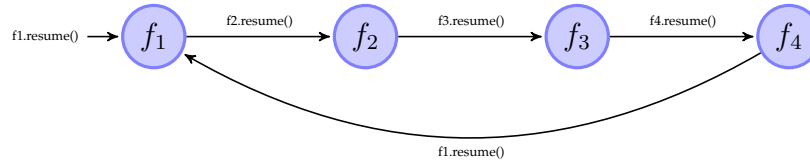


control transfer mechanism

According to the literature,⁷ coroutine-like control-transfer operations can be distinguished into the concepts of *symmetric* and *asymmetric* operations.

symmetric fiber A symmetric fiber provides a single control-transfer operation. This single operation requires that the control is passed explicitly between the fibers.

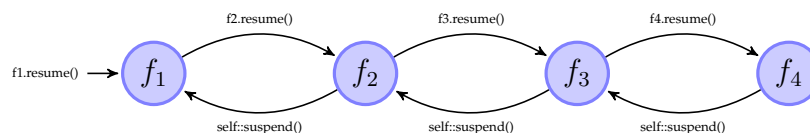


```
1 fiber_context* pf1;
2 fiber_context f4{[&pf1]{
3     pf1->resume();
4 }};
5 fiber_context f3{[&f4]{
6     f4.resume();
7 }};
8 fiber_context f2{[&f3]{
9     f3.resume();
10 }};
11 fiber_context f1{[&f2]{
12     f2.resume();
13 }};
14 pf1=&f1;
15 f1.resume();
```

In the pseudo-code example above, a chain of fibers is created.

Control is transferred to fiber f_1 at line 15 and the lambda passed to constructor of f_1 is entered. Control is transferred from fiber f_1 to f_2 at line 12 and from f_2 to f_3 (line 9) and so on. Fiber f_4 itself transfers control directly back to fiber f_1 at line 3.

asymmetric fiber Two control-transfer operations are part of asymmetric fiber's interface: one operation for resuming (`resume()`) and one for suspending (`suspend()`) the fiber. The suspending operation returns control back to the calling fiber.



```
1 // hypothetical API
2 fiber_context f4{[] {
3     self::suspend();
4 }};
5 fiber_context f3{[&f4]{
6     f4.resume();
7     self::suspend();
8 }};
9 fiber_context f2{[&f3]{
10    f3.resume();
11    self::suspend();
12 }};
13 fiber_context f1{[&f2]{
14    f2.resume();
15    self::suspend();
16 }};
```